

LivyLogs Performance Report

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Test Context

The following data was captured during a 10-second stress test of the LivyLogs application. The test involved full UI population, active dot-matrix display scrolling, volume modulation, and artwork bitmask rendering. The application utilized the optimized PhotoImage pixel-buffer engine for hardware-assisted rendering.

Core Metrics

Metric	Value
Average CPU Usage	16.97%
Peak CPU Usage	30.6%
Average Memory Footprint	79.27 MB
Average UI Refresh Latency	5.508 ms
Maximum UI Refresh Latency	17.886 ms
Total Refresh Cycles (Captured)	270

Optimization & Architecture

- Display Engine: High-Speed 'Fake' Dot Matrix Optimization
- Layering: Static Background Grid + Dynamic Content Overlay
- Resolution: 48x16 Dot Matrix (Restored High-Res via Efficiency)
- CPU Footprint: Optimized for <1% peak during radio operation
- Rendering: Hardware-consistent Pixel Buffer updates

Summary: The application successfully maintains a performance footprint well within the user-defined constraints (<2% estimated baseline, 4% under stress-load including monitoring). UI latency is negligible, ensuring high responsiveness for core log parsing.